

Jacklyn DiPietro

✉ JDIPietro05@gmail.com | in LinkedIn | 🌐 jadipietro.github.io

Summary

Ph.D. candidate in Chemistry (Penn State, June 2026) specializing in polymer coatings, surface engineering, and photothermal fabrication. Developed a laser-based patterning method for thermoset coatings with applications in fouling-release and drag reduction, with results published in RSC Applied Polymers, two manuscripts in preparation, and two pending U.S. patents.

Skills

- **Materials Characterization:** Scanning electron microscopy (SEM), optical microscopy, optical profilometry, contact angle goniometry, UV-Vis spectroscopy, FTIR spectroscopy, Raman spectroscopy, fluorescence spectroscopy, X-ray diffraction (XRD), dynamic light scattering (DLS), zeta potential analysis, tensile testing
 - **Polymer Processing and Fabrication:** Laser photothermal patterning, thermoset coatings, PDMS processing, machining (mill, lathe)
 - **Research and Operations:** Experimental design, protocol development, troubleshooting, mentorship and training, cross-functional collaboration, scientific writing
 - **Data Analysis and Visualization:** Python, IgorPro, ImageJ, Excel, Adobe Illustrator, Blender
 - **Technical Writing:** LaTeX, Microsoft Office
 - **Safety:** Laboratory safety practices, hazardous chemical handling, respirator use and PPE compliance, Class IV laser operation, laser safety protocols, standard operating procedure (SOP) development
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Education

- **Ph.D. Chemistry**, The Pennsylvania State University, State College PA *Expected June 2026*
 - **B.S. Chemistry**, West Chester University, West Chester PA *May 2022*
Double Major in Chemistry and Biochemistry, Minor in Biology
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Experience

Graduate Research Assistant, The Pennsylvania State University *January 2023 - Present*
Navy Undersea Research Program, Advisor: Benjamin Lear

Project: Photothermal patterning of polymer coatings for drag reduction and fouling-release

- Developed a novel method of generating surface features in naval coatings using **photothermal patterning**, resulting in a peer-reviewed publication[8]
- Developed standardized photothermal fabrication protocols enabling reproducible, on-demand production of patterned coating samples
- Led a cross-functional study across multiple departments evaluating fouling-release performance of patterned thermoset coatings, resulting in a manuscript in preparation
- Improved fouling-release performance, increasing the bare surface fraction after washing from 59% to 92%
- Characterized patterned polymer surfaces using **SEM, optical profilometry, Raman, FTIR, and contact angle analysis** to correlate surface morphology with improved fouling-release performance
- Mentored three undergraduate researchers in experimental design, surface characterization, and scientific writing; contributing to a manuscript in preparation

Laser Safety Officer, Lear Lab (PSU) *August 2023 - Present*

- Trained lab members on laser operation and safety
- Wrote/managed/filed laser standard operating procedures and implemented laser safety protocols to minimize exposure risks
- Maintained laser registrations, documentation and safety procedures

Graduate Teaching Assistant, The Pennsylvania State University
Courses taught: General Chemistry I lecture, General Chemistry I and II lab.

August 2022 - May 2023

- Taught undergraduate lab sessions, conducted office hours to offer additional support, and graded lab reports, homework and exams

Undergraduate Research Assistant, West Chester University

October 2020 - July 2022

Advisor: Kurt Kolasinski

Project: Mechanochemical passivation of silicon powders.

- Synthesized fluorescent porous silicon nanoparticles via HF-based etching for applications in sensors and biomedicine
 - Developed novel mechanochemical methods for the surface passivation of silicon nanoparticles, resulting in a published manuscript [9]
 - Awarded the 2021 American Chemical Society Undergraduate Award in Physical Chemistry for contribution
 - Performed characterization techniques including **FTIR**, **fluorescence spectroscopy**, **UV-Vis spectroscopy**, **dynamic light scattering**, **zeta potential**, **powder x-ray diffraction**, and **SEM**.
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Leadership

President, Gamma Sigma Epsilon Chemistry Honor Society (WCU)

August 2021 - May 2022

- Re-established local chapter at WCU, promoted the organization, managed the application process for new members, organized chapter meetings, and outreach events

President, Alchemist Club (WCU)

August 2021 - May 2022

- Organized out-reach events for the WCU chemistry department, coordinated fundraising activities, planned member meetings, tutored physical chemistry courses

Co-Organizer, WCU Alchemist Club: Young Men and Women in Charge (YMWIC) STEM Week

April 2022

- Co-organized a multi-day YMWIC STEM event for K-12 students in the Philadelphia and Chester County area, coordinated across departments and led chemistry demonstrations to promote STEM engagement
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Honors and Awards

- ARL 2025 Outstanding Graduate Student Contribution Award
 - Penn State Chemistry Second Year Graduate Award 9
 - The Nellie H. and Oscar L. Roberts Fellowship 9
 - West Chester University 1871 Award 9
 - American Chemical Society Undergraduate Award in Physical Chemistry 9
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Publications and Patents

- **J.A. DiPietro**, K.W. Kolasinski. Characterization of Mechanochemical Modification of Porous Silicon with Arginine. *Surfaces* 2022, 5, 143-154. 9
- **J.A. DiPietro**, B. Ludwig, B. Lear. Photothermal Patterning of Polydimethylsiloxane. *RSC Applied Polymers* 2025 9
- **J.A. DiPietro**, B.E. Sidore, et al., Shark-Skin Inspired Photothermal Patterning For Fouling-Release Applications, *in preparation*
- B.E. Sidore, **J.A. DiPietro**, et al., Wear Behavior of Photothermally Generated Bio-Inspired Patterns in Polydimethylsiloxane, *in preparation*
- U.S. Provisional Patent Application No. 63/736,637. One-step curing and micron-scale patterning of polymers using photothermal curing of multilayer multicomponent films *pending*
- U.S. Provisional Patent Application No. 63/620,944. Photothermal curing of polymers on thermally sensitive substrates *pending*